

Python Operators – 10 Practice Tasks: Answers

Task 1 — Arithmetic Operators

```
a = 25
b = 4
print(a + b) # 29
print(a - b) # 21
print(a * b) # 100
print(a / b) # 6.25
print(a // b) # 6
print(a % b) # 1
print(a ** b) # 390625
```

Task 2 — Assignment Operators

```
x = 10
x += 5
print(x) # 15
x -= 3
print(x) # 12
x *= 2
print(x) # 24
x /= 4
print(x) # 6.0
x %= 3
print(x) # 0.0
```

Task 3 — Comparison Operators

```
a = 15
b = 10
print(a > b) # True
print(a < b) # False
print(a >= b) # True
print(a <= b) # False
print(a == b) # False
print(a != b) # True
```

Task 4 — Logical Operators

```
age = 25
salary = 50000
print(age > 18 and salary > 30000) # True
print(age < 18 or salary > 30000) # True
print(not(age > 18)) # False
```

Task 5 — Identity Operators

```
a = [10, 20]
b = a
c = [10, 20]

print(a is b) # True
print(a is c) # False
print(b is c) # False
print(a == b) # True
print(a == c) # True
```

Explanation:
'is' checks whether two variables refer to the same object.
'==' checks whether their values are equal. a and c contain the same values but are separate list objects, so a is c is False while a == c is True.

Task 6 — Membership Operators

```
numbers = [10, 20, 30, 40, 50]
print(20 in numbers) # True
```

```

print(60 in numbers)      # False
print(30 not in numbers)  # False
print(100 not in numbers) # True

```

Task 7 — Bitwise Operators

```

a = 10  # binary: 1010
b = 6   # binary: 0110

print(a & b)  # 2
print(a | b)  # 14
print(a ^ b)  # 12
print(~a)     # -11
print(a << 1) # 20
print(a >> 1) # 5

```

Task 8 — Predict the Output

```

a = 10
b = 20

print(a + b)      # 30
print(a > b)      # False
print(a < b)      # True
print(a == b)     # False
print(a != b)     # True
print(a < b and b > 15) # True
print(a > b or b > 15)  # True
print(not(a > b))     # True

```

Task 9 — Mixed Operators

```

a = 10
b = 5
c = 10

print(a + b * 2)      # 20    -> arithmetic
print(a > b and c == a) # True  -> comparison + logical
print(a == c or b > c) # True  -> comparison + logical
print(a is c)         # True  -> identity (small integers are commonly interned)
print(a in [5, 10, 15]) # True -> membership
print(a & b)          # 0     -> bitwise

```

Task 10 — Real Challenge

```

a = 20
b = 10
numbers = [10, 20, 30]
x = a

# Arithmetic
print(a + b)

# Assignment
x += b
print(x)

# Comparison
print(a > b)

# Logical
print(a > b and b < a)

# Identity
print(x is a)

# Membership
print(a in numbers)

# Bitwise
print(a & b)

```